Software Requirements Specification

E-Commerce Platform

1. Functional Requirements (49)

Format: Actor | Verb | Object. Row color: Blue = Customer · Green = Seller · Orange = Administrator · Purple = System

#	Actor	Verb	Object	Category	
1	Customer	shall search	products using keywords, categories, and filters	Functional: Customer	Elda Toska
2	Customer	shall browse	products by category and subcategory	Functional: Customer	Elda Toska
3	Customer	shall filter	product listings by price, rating, and availability	Functional: Customer	Elda Toska
4	Customer	shall sort	product listings by price, popularity, newest, and rating	Functional: Customer	Elda Toska
5	Customer	shall compare	multiple products based on price, specifications, ratings, and availability	Functional: Customer	Elda Toska
6	Customer	shall upload	an image to search for visually similar products in the catalog	Functional: Customer	Elda Toska
7	Customer	shall view	a list of recently viewed products	Functional: Customer	Elda Toska
8	Customer	shall add	products to the shopping cart	Functional: Customer	Sindi Terihati
9	Customer	shall update	product quantities in the shopping cart	Functional: Customer	Sindi Terihati
10	Customer	shall remove	products from the shopping cart before checkout	Functional: Customer	Sindi Terihati
11	Customer	shall add	products to a wishlist for future purchases	Functional: Customer	Sindi Terihati
12	Customer	shall save	a wishlist for future reference	Functional: Customer	Sindi Terihati
13	Customer	shall place	orders and complete purchases using available payment methods	Functional: Customer	Sindi Terihati
14	Customer	shall cancel	orders before shipment	Functional: Customer	Sindi Terihati
15	Customer	shall submit	refund requests for eligible purchases	Functional: Customer	Elisa Spaho

#	Actor	Verb	Object	Category	
16	Customer	shall track	the real-time location of their delivery via live map updates	Functional: Customer	Elisa Spaho
17	Customer	shall receive	notifications for order confirmation, shipping updates, and delivery status	Functional: Customer	Elisa Spaho
18	Customer	shall view	an estimated delivery time based on shipping method and delivery location	Functional: Customer	Elisa Spaho
19	Customer	shall download	auto-generated invoices or receipts for each completed purchase	Functional: Customer	Elisa Spaho
20	Customer	shall subscribe	to back-in-stock notifications for out-of-stock products	Functional: Customer	Elisa Spaho
21	Customer	shall receive	alerts when the price of a saved or viewed product decreases	Functional: Customer	Elisa Spaho
22	Customer	shall save	multiple payment methods (credit cards, digital wallets) in their account	Functional: Customer	Stilianos Stroka
23	Customer	shall add	multiple delivery addresses to their account	Functional: Customer	Stilianos Stroka
24	Customer	shall edit	saved delivery addresses in their account	Functional: Customer	Stilianos Stroka
25	Customer	shall delete	saved delivery addresses from their account	Functional: Customer	Stilianos Stroka
26	Customer	shall select	a preferred delivery address during checkout	Functional: Customer	Stilianos Stroka
27	Customer	shall submit	reviews and ratings for purchased products	Functional: Customer	Stilianos Stroka
28	Customer	shall view	personalized product recommendations based on browsing and purchase history	Functional: Customer	Stilianos Stroka
29	Customer	shall submit	support tickets to report issues related to orders, payments, or products	Functional: Customer	Ederta Kariqi
30	Customer	shall view	the carbon footprint report for their selected shipping method	Functional: Customer	Ederta Kariqi
31	Customer	shall view	video-based unboxing and product reviews before making a purchase	Functional: Customer	Ederta Kariqi
32	Seller	shall add	product listings including descriptions, prices, and images	Functional: Seller	Ederta Kariqi
33	Seller	shall manage	existing product listings (edit, update, remove)	Functional: Seller	Ederta Kariqi
34	Seller	shall create	product bundles combining multiple products at a combined price	Functional: Seller	Ederta Kariqi

#	Actor	Verb	Object	Category	
35	Seller	shall view	sales reports and analytics (total sales, revenue, product performance)	Functional: Seller	Ederta Kariqi
36	Seller	shall monitor	stock levels and receive alerts when inventory reaches low-stock thresholds	Functional: Seller	Asja Xhelezi
37	Seller	shall analyze	predictive inventory analytics to forecast trend-based restocking needs	Functional: Seller	Asja Xhelezi
38	Seller	shall utilize	the AI Price Optimizer to adjust product prices for market positioning	Functional: Seller	Asja Xhelezi
39	Administrator	shall monitor	customer orders and platform inventory	Functional: Administrator	Asja Xhelezi
40	Administrator	shall manage	customer orders and platform inventory	Functional: Administrator	Asja Xhelezi
41	Administrator	shall create	promotional offers, discount codes, and seasonal sales campaigns	Functional: Administrator	Asja Xhelezi
42	Administrator	shall manage	promotional offers, discount codes, and seasonal sales campaigns	Functional: Administrator	Asja Xhelezi
43	System	shall generate	personalized product recommendations based on browsing history, purchase behavior, and ratings	Functional: System	Arjola Metaliaj
44	System	shall display	customer reviews and ratings on product pages	Functional: System	Arjola Metaliaj
45	System	shall display	estimated delivery time for each order	Functional: System	Arjola Metaliaj
46	System	shall validate	transaction details and other order data before final confirmation	Functional: System	Arjola Metaliaj
47	System	shall generate	downloadable invoices or receipts for completed purchases	Functional: System	Arjola Metaliaj
48	System	shall generate	administrative platform analytics and sales performance	Functional: System	Arjola Metaliaj
49	System	shall update	product inventory in seller dashboards and customer storefronts	Functional: System	Arjola Metaliaj

2. Non-Functional Requirements (35)

Lecture Categorization based on Sommerville's NFR taxonomy (Product / Organizational / External). Row color reflects lecture category type.

Requirement Categorization Legend

- Product – Security Requirements
- Product – Efficiency – Performance Requirements

- Product – Efficiency – Usability Requirements
- Product – Dependability Requirements
- Organizational – Operational / Environmental Requirements
- Organizational – Development Requirements
- External – Regulatory / Legislative Requirements

#	Requirement	Category	Lecture Categorization	
1	The system shall ensure secure processing of payment transactions and user authentication.	Non-Functional: Security	Product – Security Requirements	Elda Toska
2	The system shall protect customer personal data and purchase history from unauthorized access.	Non-Functional: Privacy	External – Regulatory / Legislative Requirements	Elda Toska
3	The system shall display product pages and search results within an acceptable response time.	Non-Functional: Performance	Product – Efficiency – Performance Requirements	Elda Toska
4	The system shall ensure that all product images and video reviews load within 3 seconds on mobile devices.	Non-Functional: Performance (Efficiency)	Product – Efficiency – Performance Requirements	Elda Toska
5	The system shall support increasing numbers of users, products, and transactions without significant degradation in performance.	Non-Functional: Scalability	Organizational – Operational Requirements	Elda Toska
6	The system shall maintain at least 99% uptime to ensure continuous access for customers, sellers, and administrators.	Non-Functional: Availability	Product – Dependability Requirements	Sindi Terihati
7	The system shall perform regular automated backups of critical data and support recovery in case of system failure.	Non-Functional: Reliability	Product – Dependability Requirements	Sindi Terihati
8	The system shall be able to restore critical services and data within a defined recovery time in case of system failure.	Non-Functional: Disaster Recovery	Product – Dependability Requirements	Sindi Terihati
9	The system shall provide a clear and intuitive user interface that allows users to complete common shopping tasks easily with minimal training.	Non-Functional: Usability	Product – Efficiency – Usability Requirements	Sindi Terihati
10	The system shall be compatible with major web browsers and mobile devices to ensure accessibility across different platforms.	Non-Functional: Compatibility	Organizational – Environmental Requirements	Sindi Terihati

#	Requirement	Category	Lecture Categorization	
11	The system shall maintain detailed logs of user actions, transactions, and administrative changes to support auditing, troubleshooting, and regulatory compliance.	Non-Functional: Auditability	External – Regulatory Requirements	Elisa Spaho
12	The system shall generate detailed error logs for every failed API call to external logistics and payment providers to facilitate rapid troubleshooting.	Non-Functional: Maintainability	Organizational – Development Requirements	Elisa Spaho
13	The system shall interface seamlessly with external logistics providers' APIs for real-time data sync.	Non-Functional: Interoperability	External – Regulatory / Legislative Requirements	Elisa Spaho
14	The system shall ensure that product data, order records, and inventory information remain consistent across all system modules.	Non-Functional: Data Consistency	Product – Dependability Requirements	Elisa Spaho
15	The system shall encrypt all sensitive user data (passwords, payment details) at rest and in transit using industry-standard encryption protocols.	Non-Functional: Security	Product – Security Requirements	Elisa Spaho
16	The system shall enforce role-based access control to restrict system functions to authorized users (customers, sellers, administrators).	Non-Functional: Security	Product – Security Requirements	Stilianos Troka
17	The system shall comply with relevant data protection regulations (e.g., GDPR) for the storage and processing of customer personal data.	Non-Functional: Privacy	External – Legislative Requirements	Stilianos Troka
18	The system shall process checkout transactions and payment confirmations within 5 seconds under normal load conditions.	Non-Functional: Performance	Product – Efficiency – Performance Requirements	Stilianos Troka
19	The system shall support horizontal scaling to handle peak traffic loads such as flash sales without degradation in response time.	Non-Functional: Scalability	Organizational – Operational Requirements	Stilianos Troka
20	The system shall deliver real-time order status notifications to customers within 30 seconds of a status change.	Non-Functional: Performance	Product – Efficiency – Performance Requirements	Stilianos Troka
21	The system shall maintain a mean time between failures (MTBF) that	Non-Functional: Reliability	Product – Dependability Requirements	Ederta Kariqi

#	Requirement	Category	Lecture Categorization	
	supports continuous platform operation for customers and sellers.			
22	The system shall support graceful degradation so that core shopping and checkout functions remain operational even if ancillary services fail.	Non-Functional: Availability	Product – Dependability Requirements	Ederta Kariqi
23	The system shall provide accessible user interfaces conforming to WCAG 2.1 AA standards to support users with disabilities.	Non-Functional: Usability	Product – Efficiency – Usability Requirements	Ederta Kariqi
24	The system shall display meaningful error messages to users when operations fail, without exposing internal system details.	Non-Functional: Usability	Product – Efficiency – Usability Requirements	Ederta Kariqi
25	The system shall be deployable across major cloud infrastructure providers to avoid vendor lock-in.	Non-Functional: Compatibility	Organizational – Environmental Requirements	Ederta Kariqi
26	The system shall provide a documented and versioned REST API to facilitate integration with third-party services and seller tools.	Non-Functional: Interoperability	External – Legislative / Regulatory Requirements	Asja Xhelezi
27	The system shall allow new features and modules to be deployed without requiring full system downtime.	Non-Functional: Maintainability	Organizational – Development Requirements	Asja Xhelezi
28	The system codebase shall follow modular architecture principles to facilitate independent testing, updating, and replacement of components.	Non-Functional: Maintainability	Organizational – Development Requirements	Asja Xhelezi
29	The system shall support multi-currency and multi-language configurations to enable international market expansion.	Non-Functional: Compatibility	Organizational – Operational Requirements	Asja Xhelezi
30	The system shall pass automated security vulnerability scans prior to each production release.	Non-Functional: Security	Product – Security Requirements	Asja Xhelezi
31	The system shall log all access to sensitive customer data including who accessed it, when, and from where.	Non-Functional: Auditability	External – Regulatory Requirements	Arjola Metaliaj
32	The system shall provide an administrative dashboard for monitoring system health, uptime, and performance metrics in real time.	Non-Functional: Availability	Organizational – Operational Requirements	Arjola Metaliaj

#	Requirement	Category	Lecture Categorization	
33	The system shall retain transaction records and audit logs for a minimum period defined by applicable regulatory requirements.	Non-Functional: Auditability	External – Regulatory / Legislative Requirements	Arjola Metaliaj
34	The system shall support automated regression testing pipelines to validate system behavior after each deployment.	Non-Functional: Maintainability	Organizational – Development Requirements	Arjola Metaliaj
35	The system shall enforce session timeout policies and secure session management to protect user accounts from unauthorized access.	Non-Functional: Security	Product – Security Requirements	Arjola Metaliaj